

Zewail City of Science and Technology

Nanotechnology and Nanoelectronics Engineering Program



NANENG 325 ASIC & FPGA Design

32-Bit MAC Unit Using Booth Encoding, a Dadda tree and CSP³L Flip Flops

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Post-layout Verification

Having laid out the whole MAC unit and its blocks and proved their functionality pre-layout, we now have to lay-out these blocks and prove they function well after layout.

1. XOR Gate

All the XOR gates used in the project have been implemented using Transmission gates to save area and enhance delay and performance.

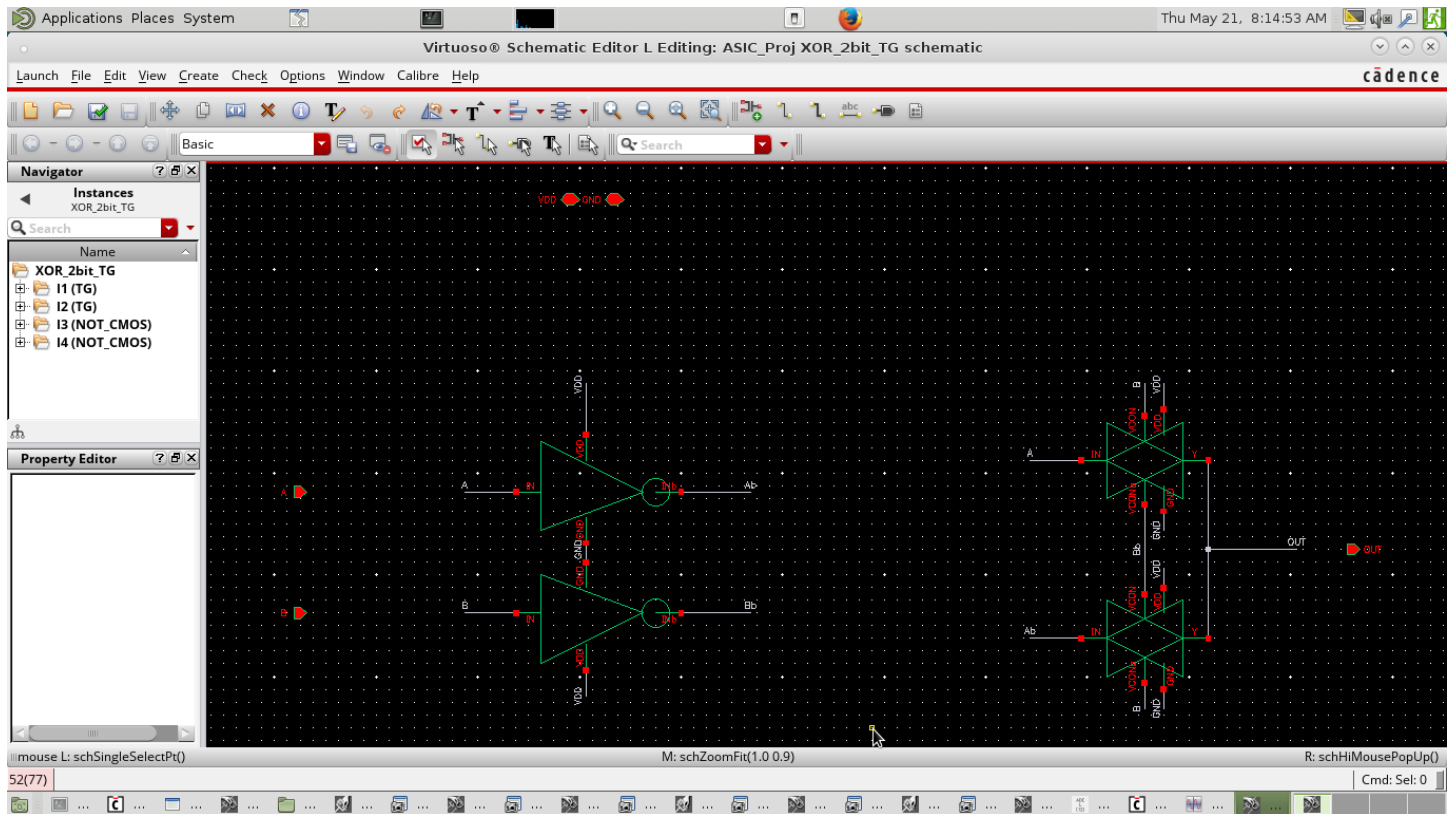


Figure 1: XOR Gate schematic.

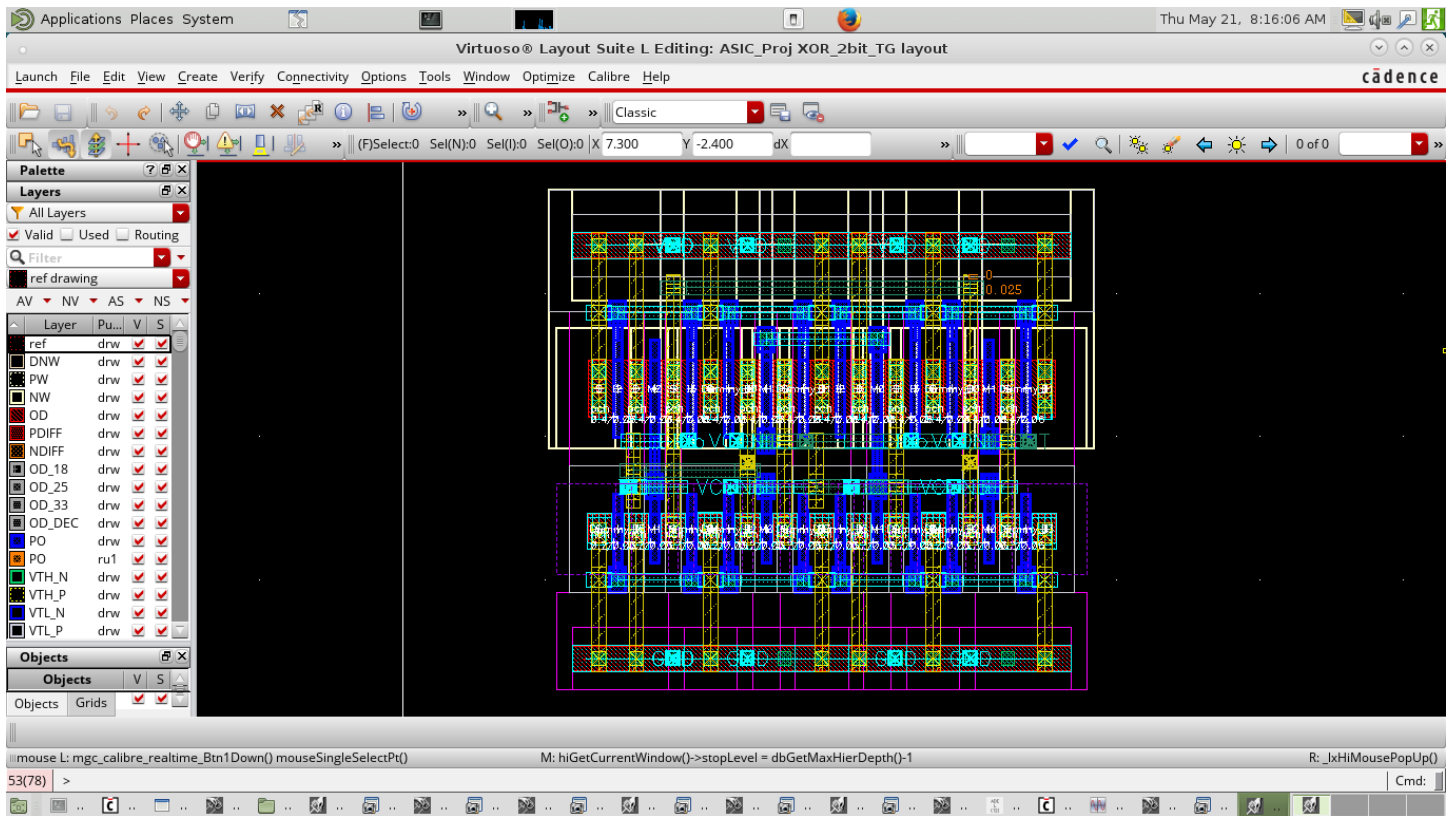


Figure 2: XOR Gate layout.

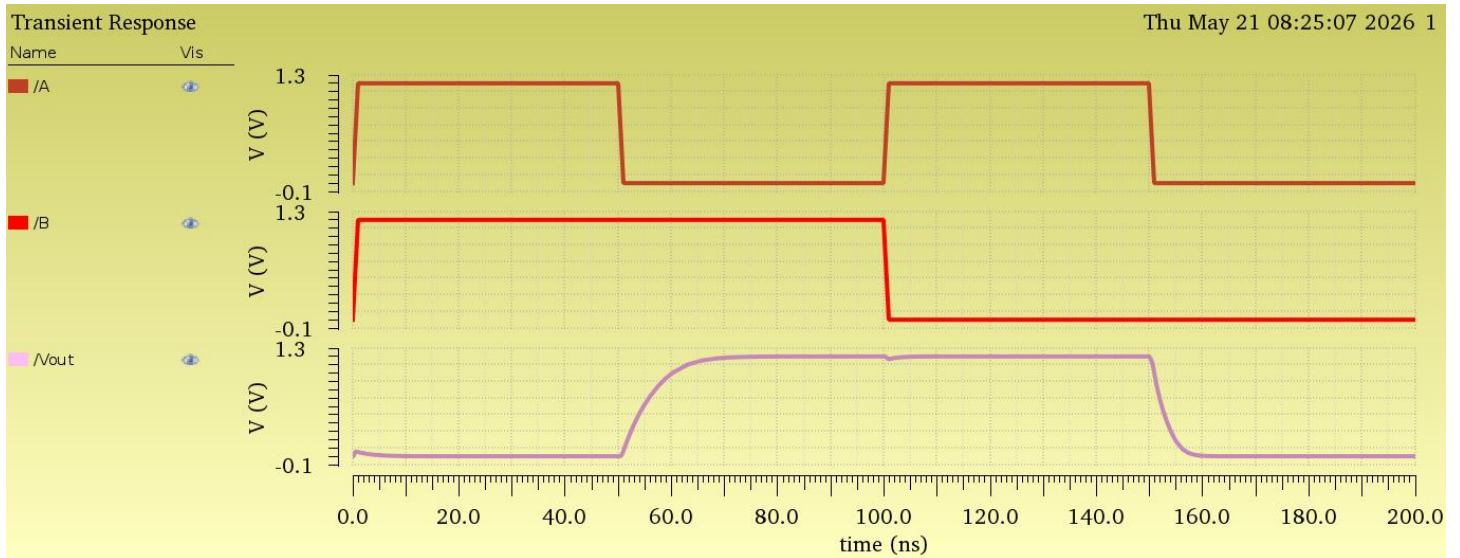


Figure 3: XOR Gate Test Bench Pre-Layout.

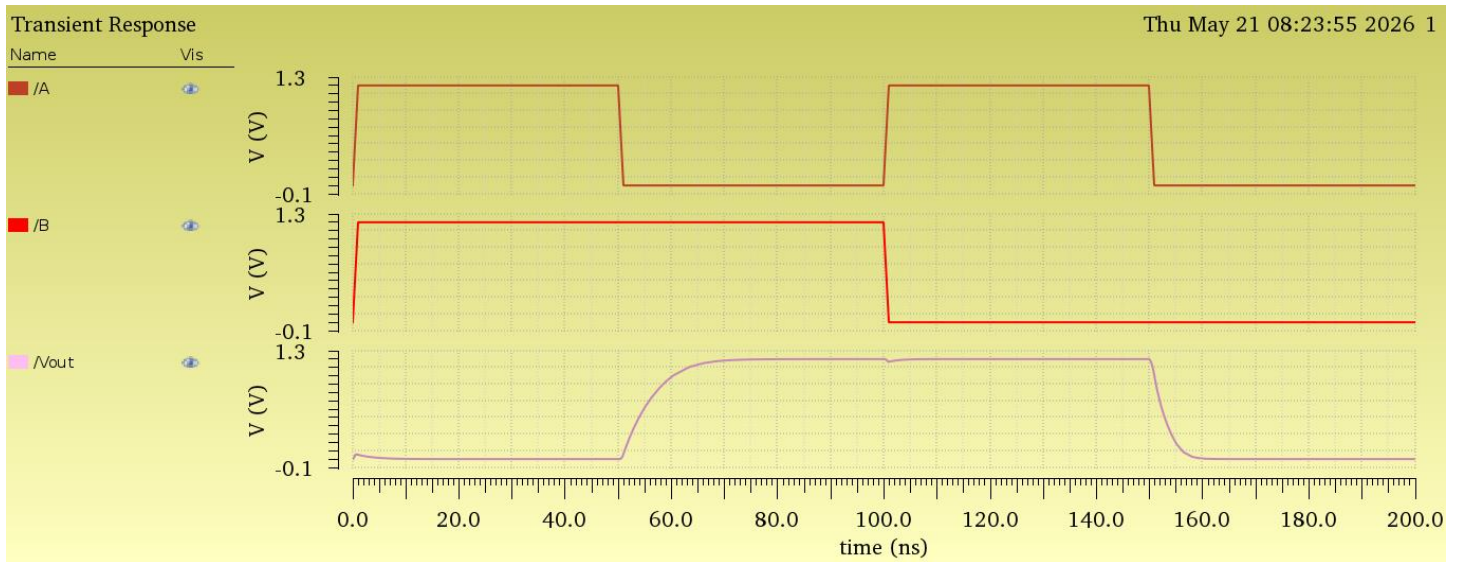


Figure 4: XOR Gate Test Bench Post Layout.

As we observe, the behaviour is identical pre- and post-layout.

2. Half Adder

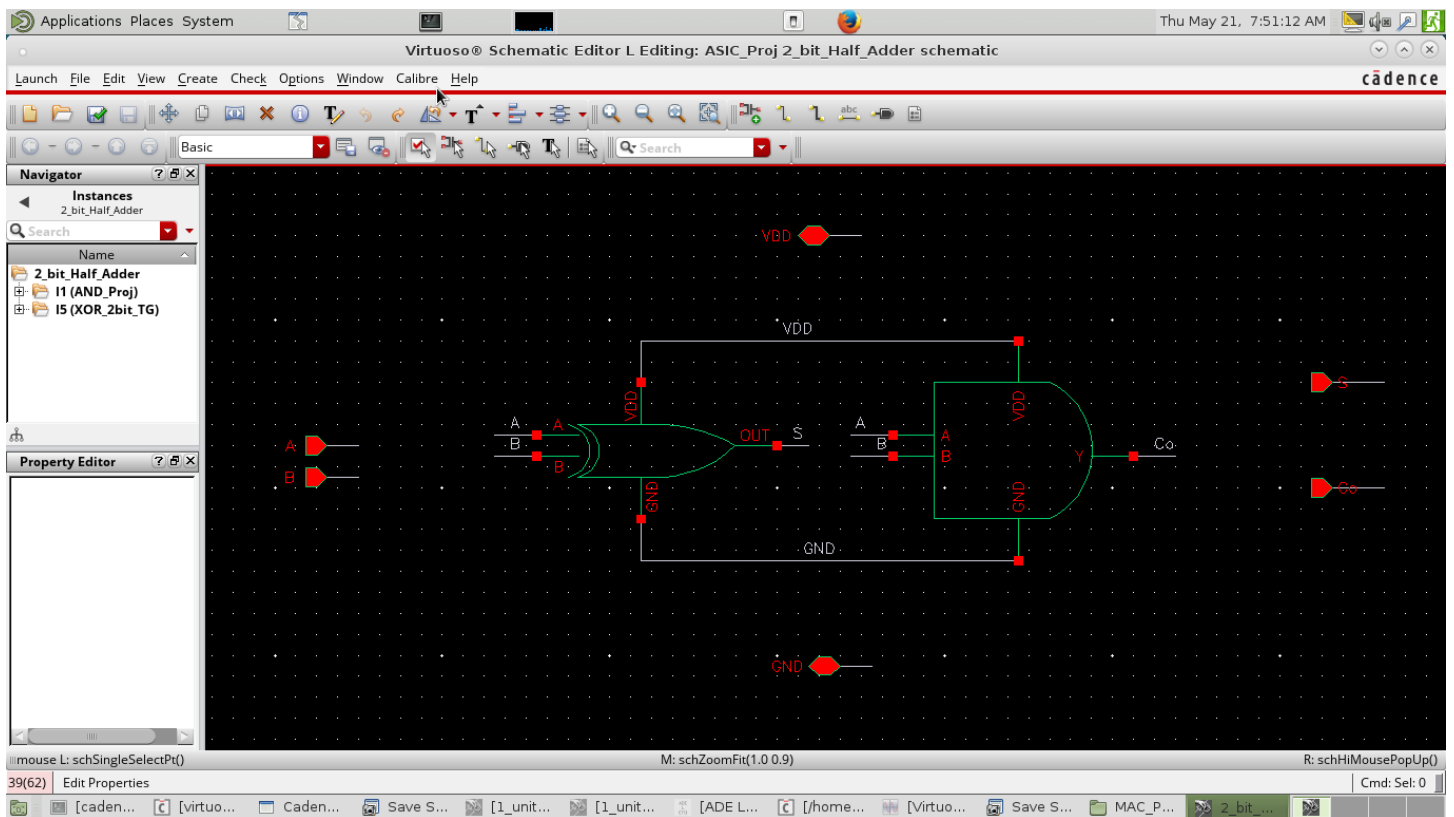


Figure 5: Half Adder Schematic.

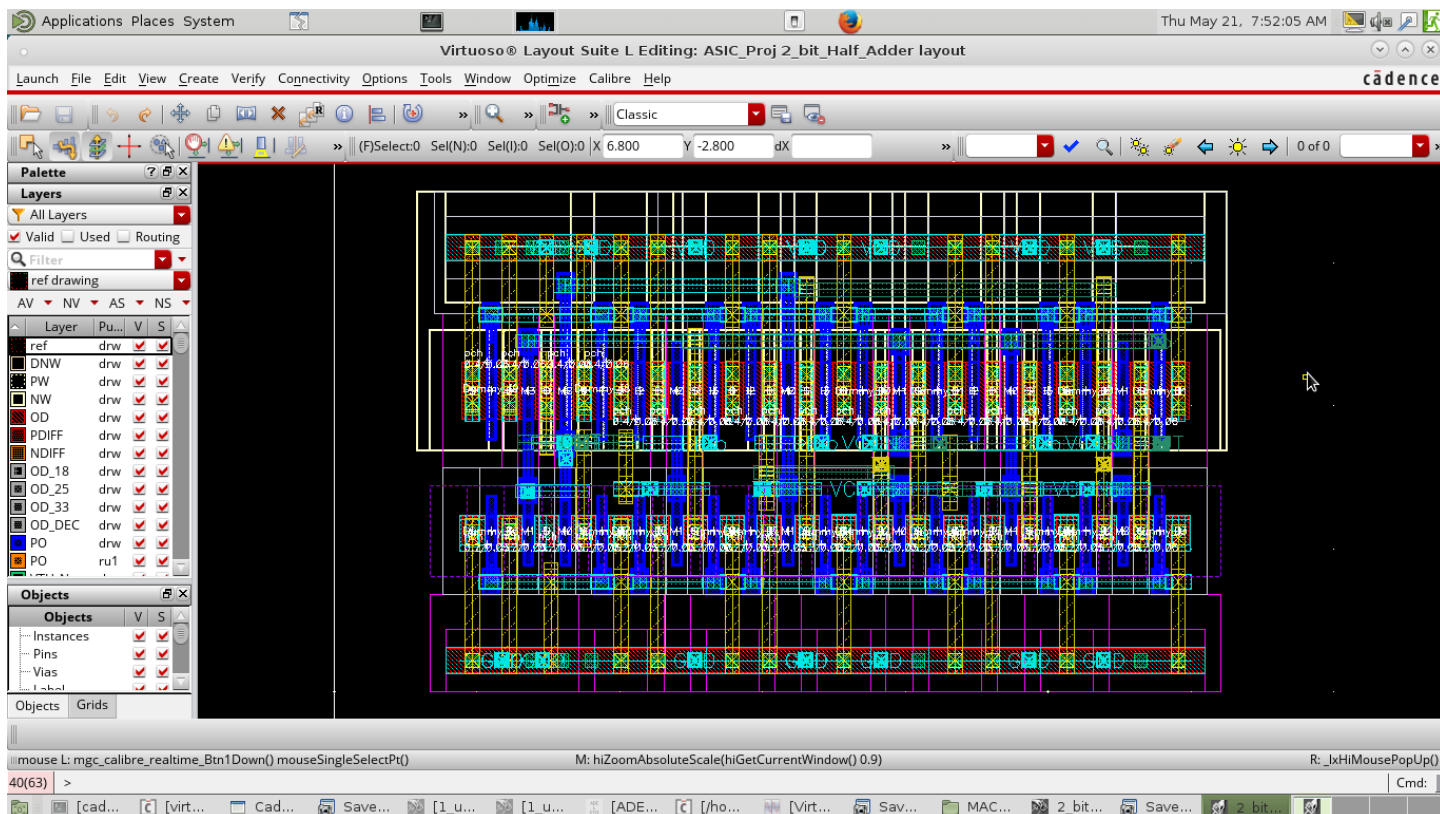


Figure 6: Half Adder layout.

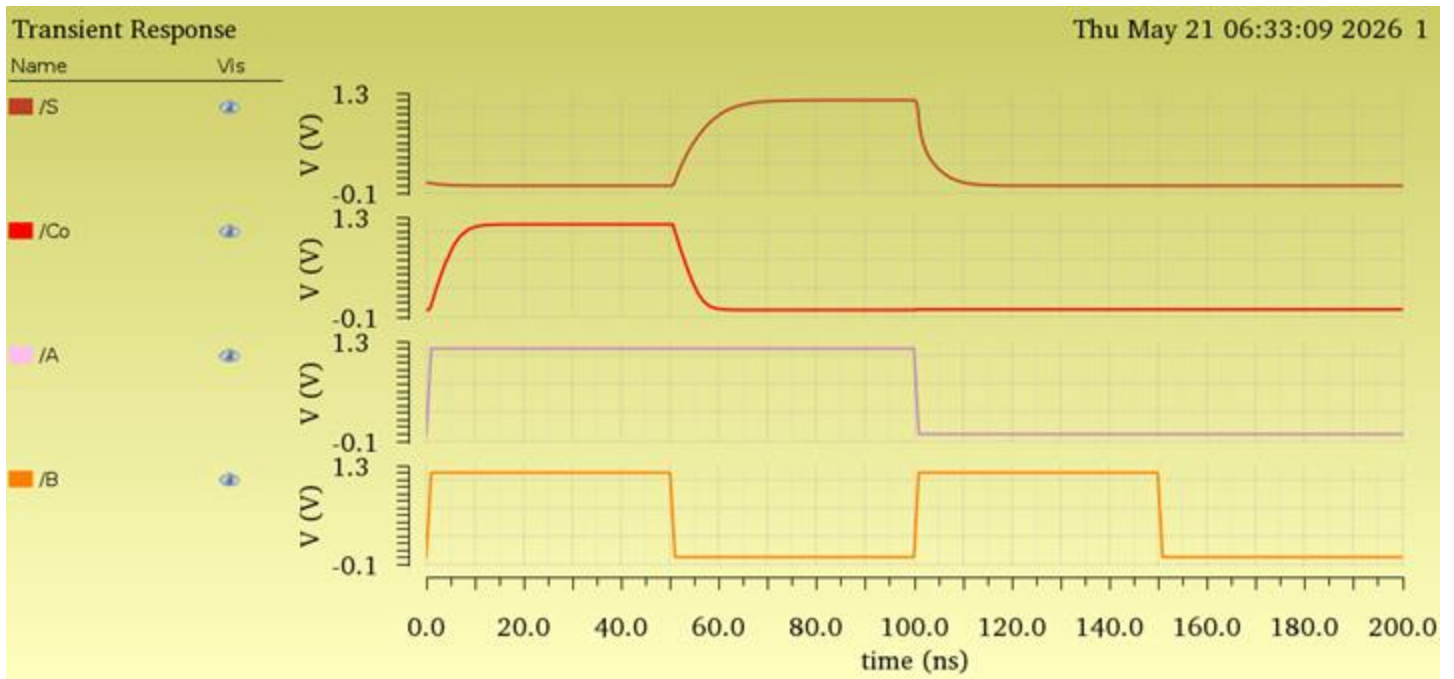


Figure 7: Half Adder Test Bench pre-layout.

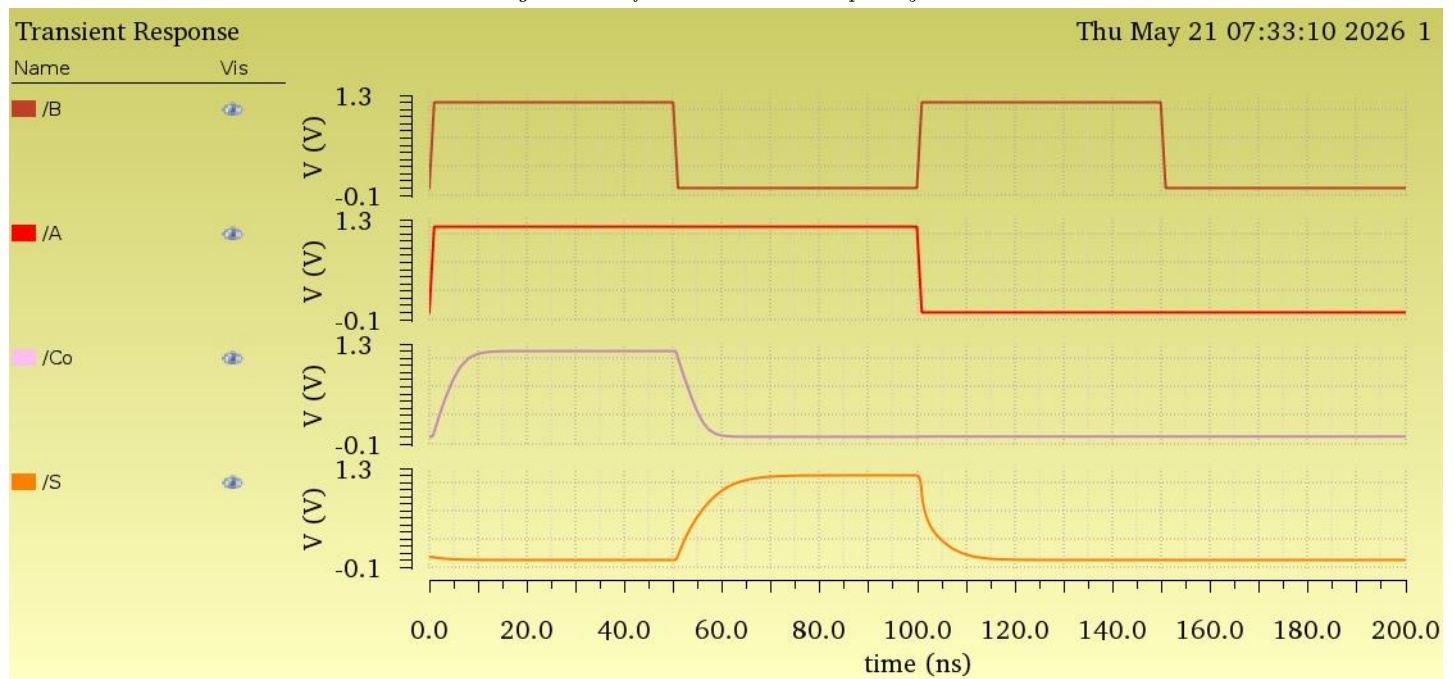


Figure 8: Half Adder Test Bench pre-layout

The behaviour is identical pre- and post-layout.

3. Mirror Full Adder

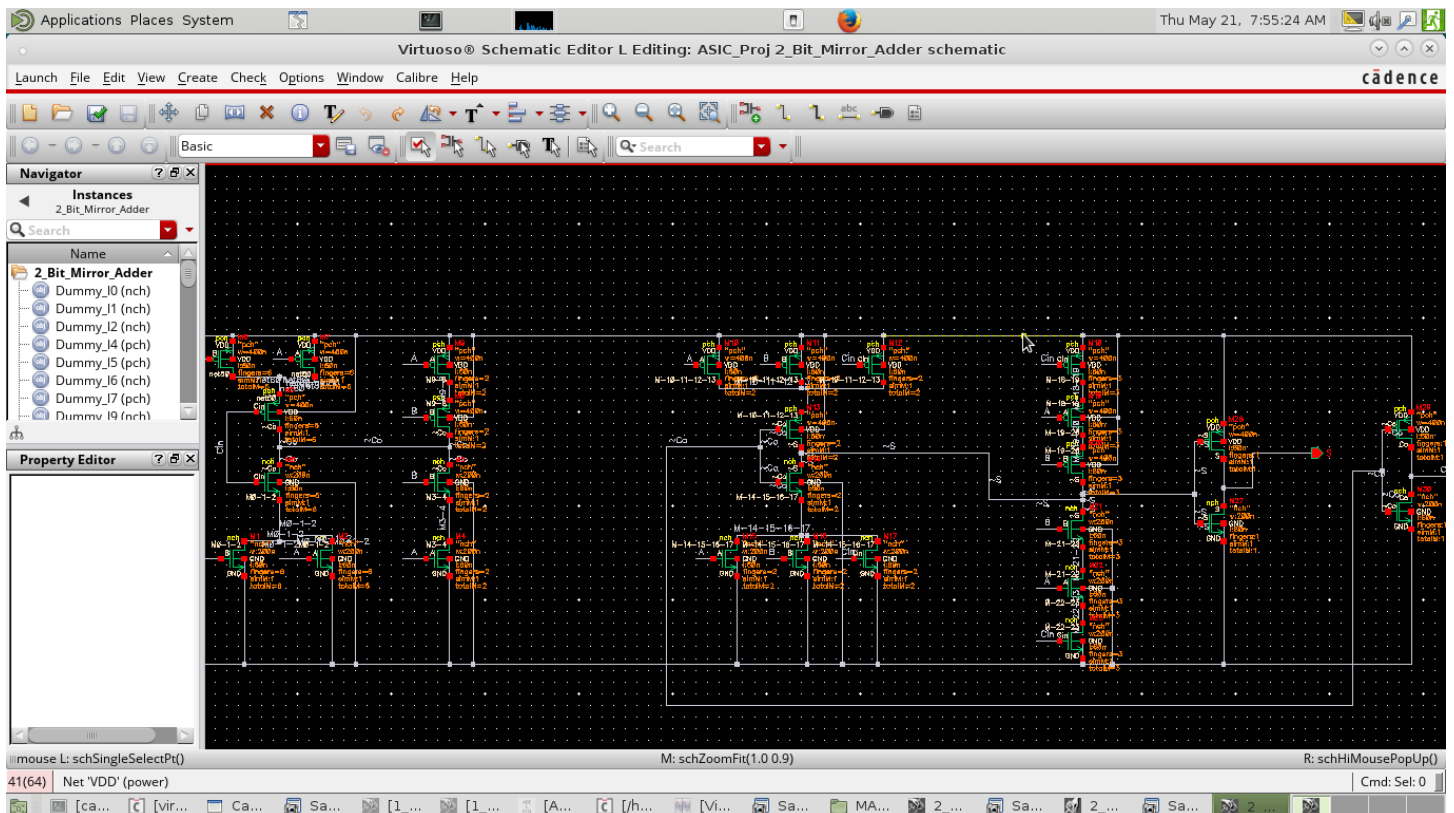


Figure 9: Mirror Full Adder Schematic.

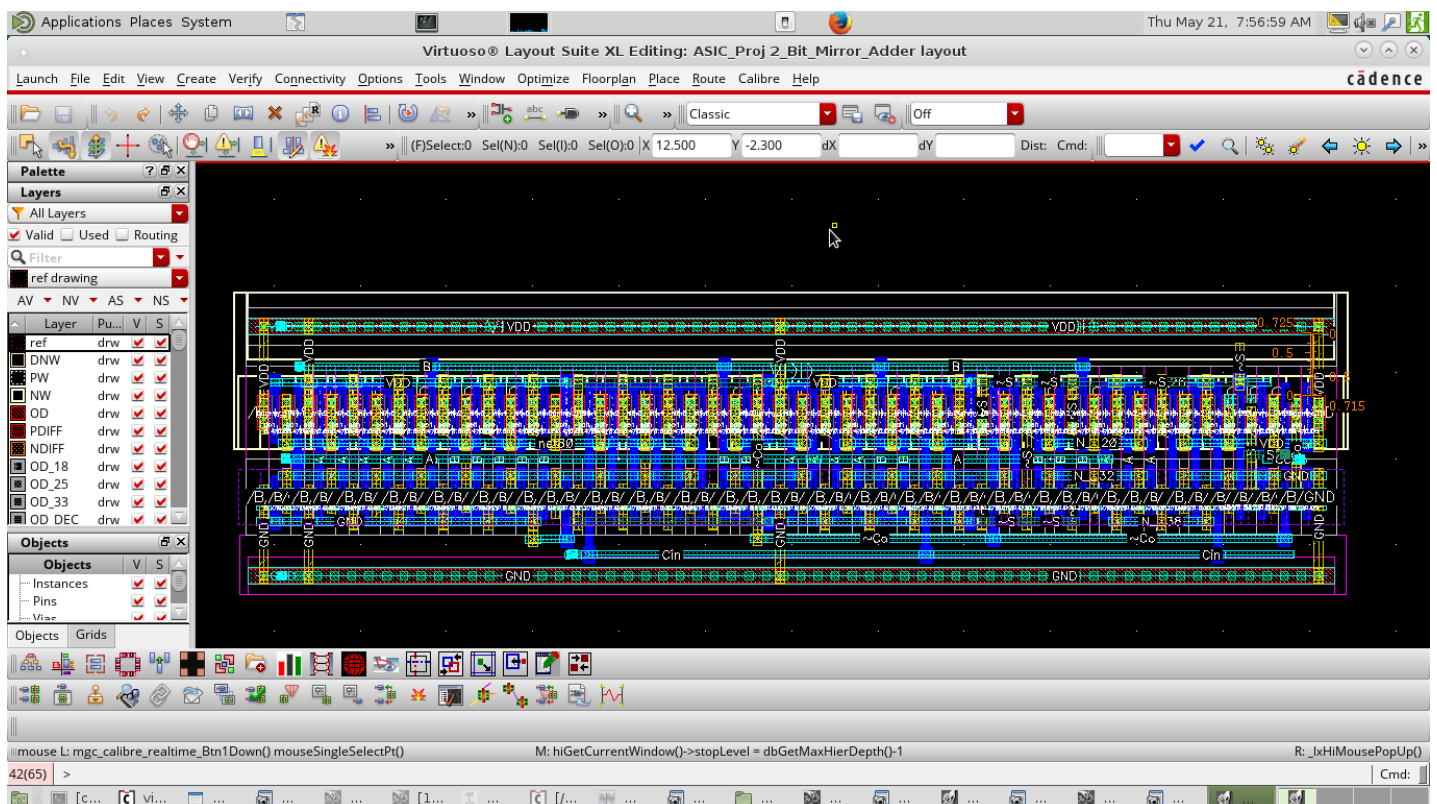


Figure 10: Mirror Full Adder Layout.

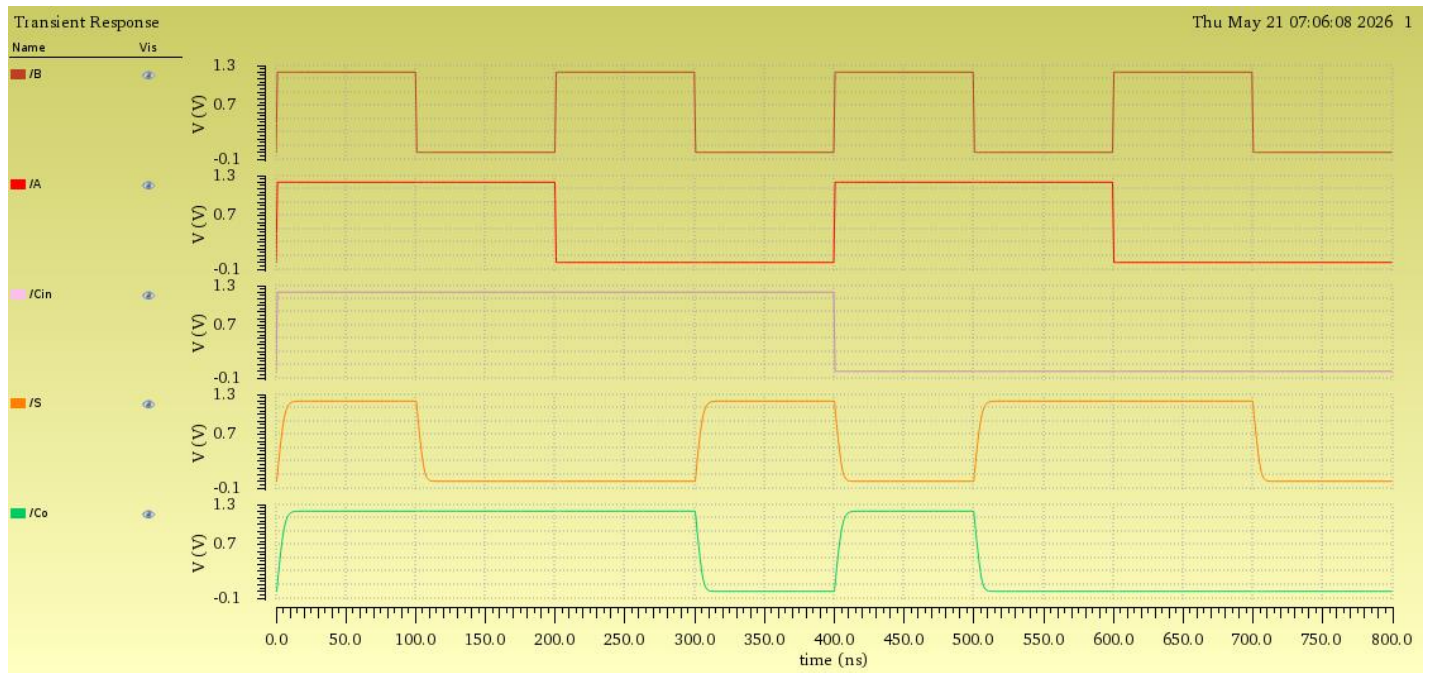


Figure 11: Full adder Test Bench pre-layout.

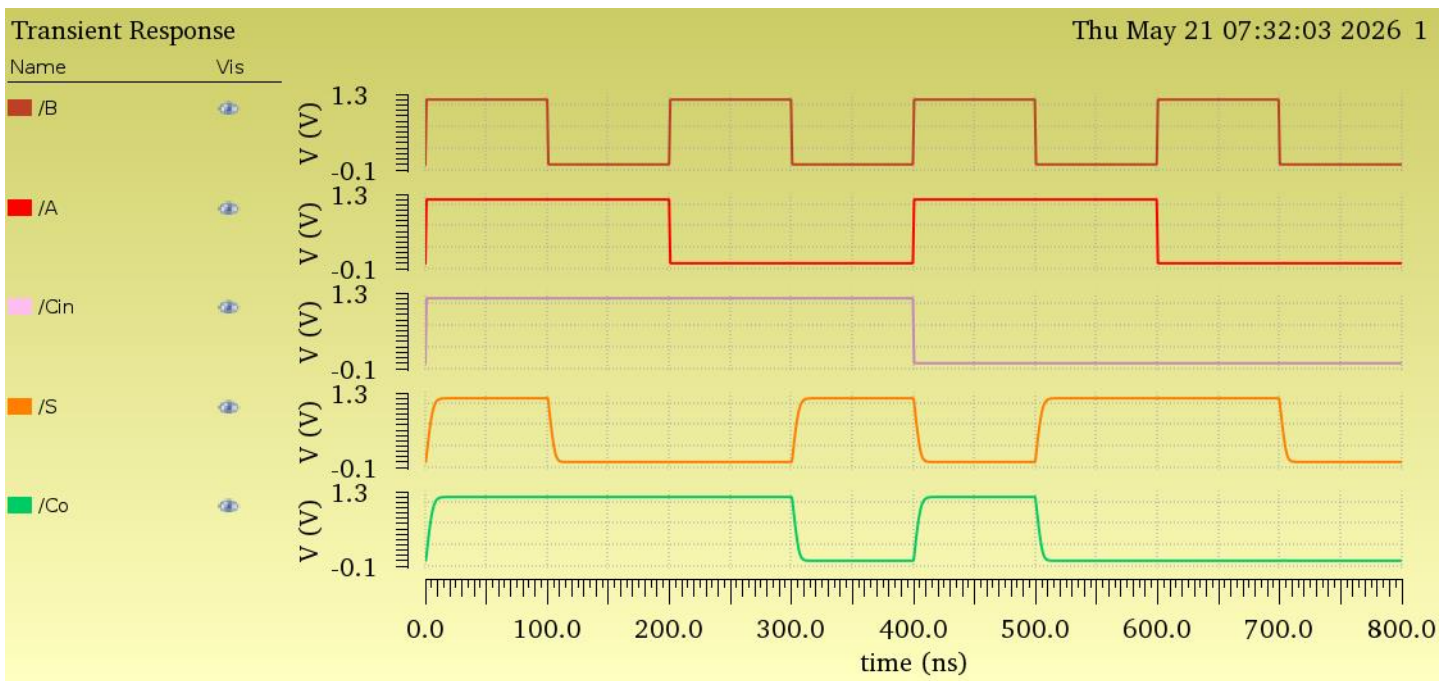


Figure 12: Full adder Test Bench post-layout.

As we observe, the behaviour is identical pre- and post-layout.

4. Booth Encoder

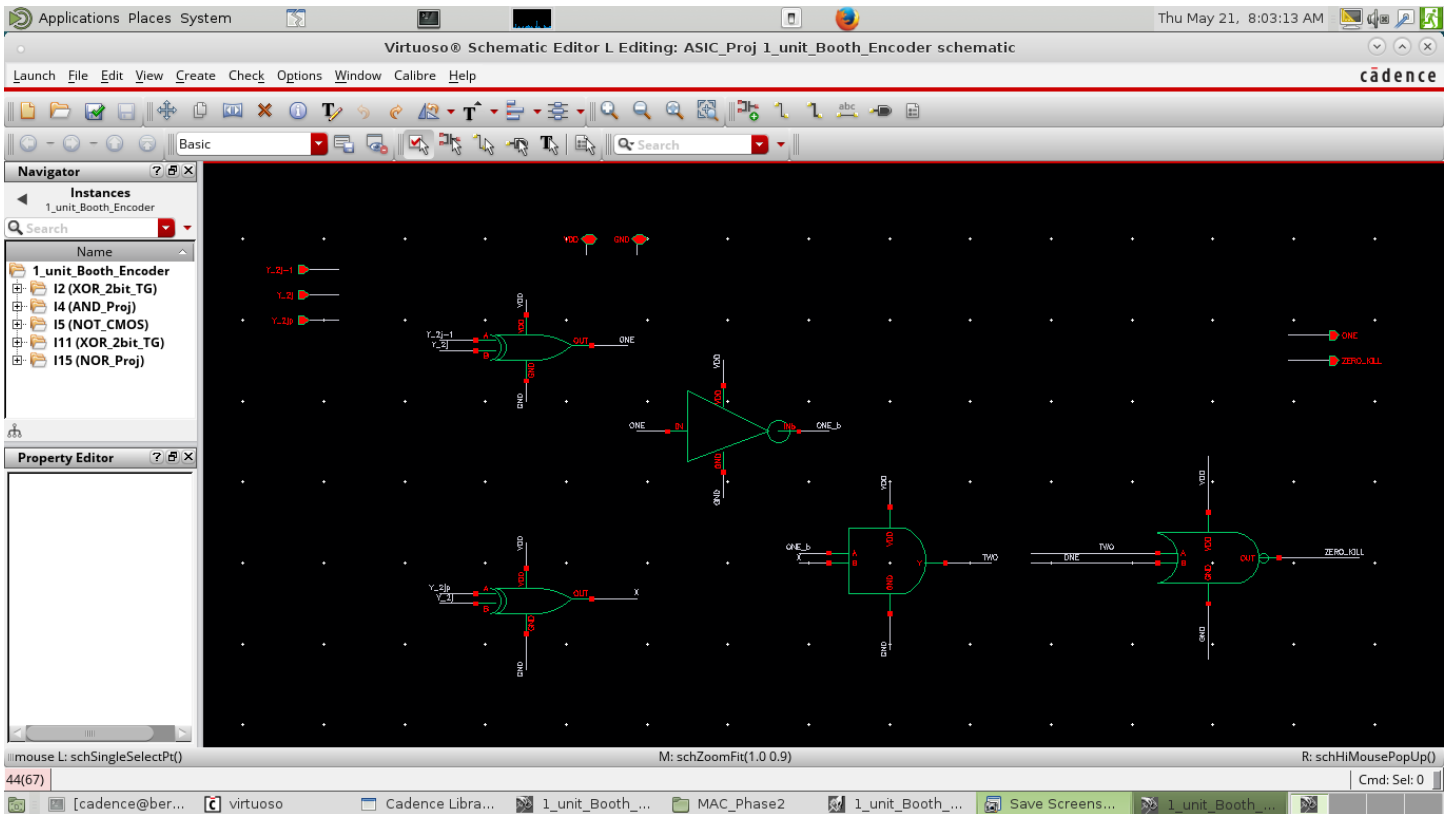


Figure 13: Booth Encoder Schematic.

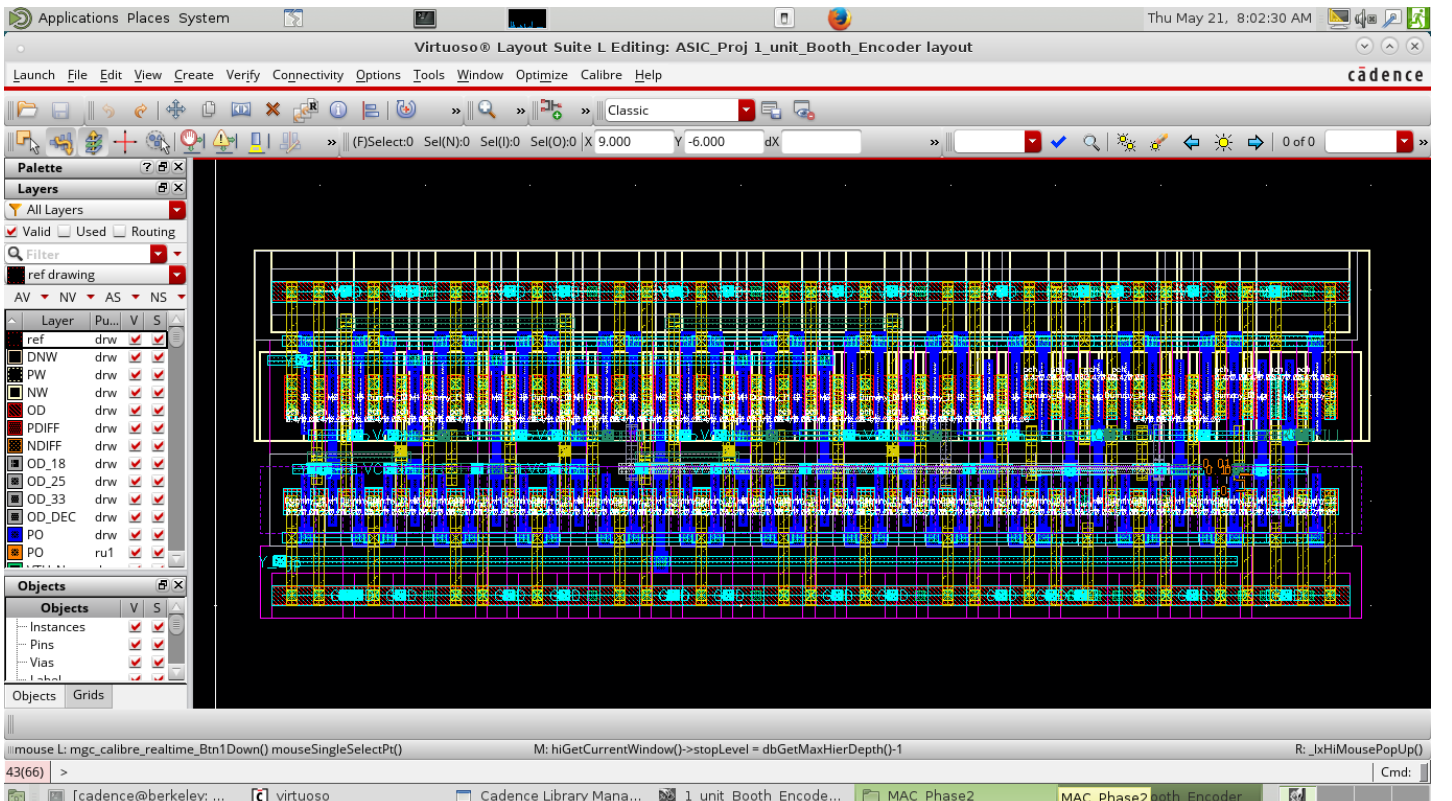


Figure 14: Booth Encoder layout.

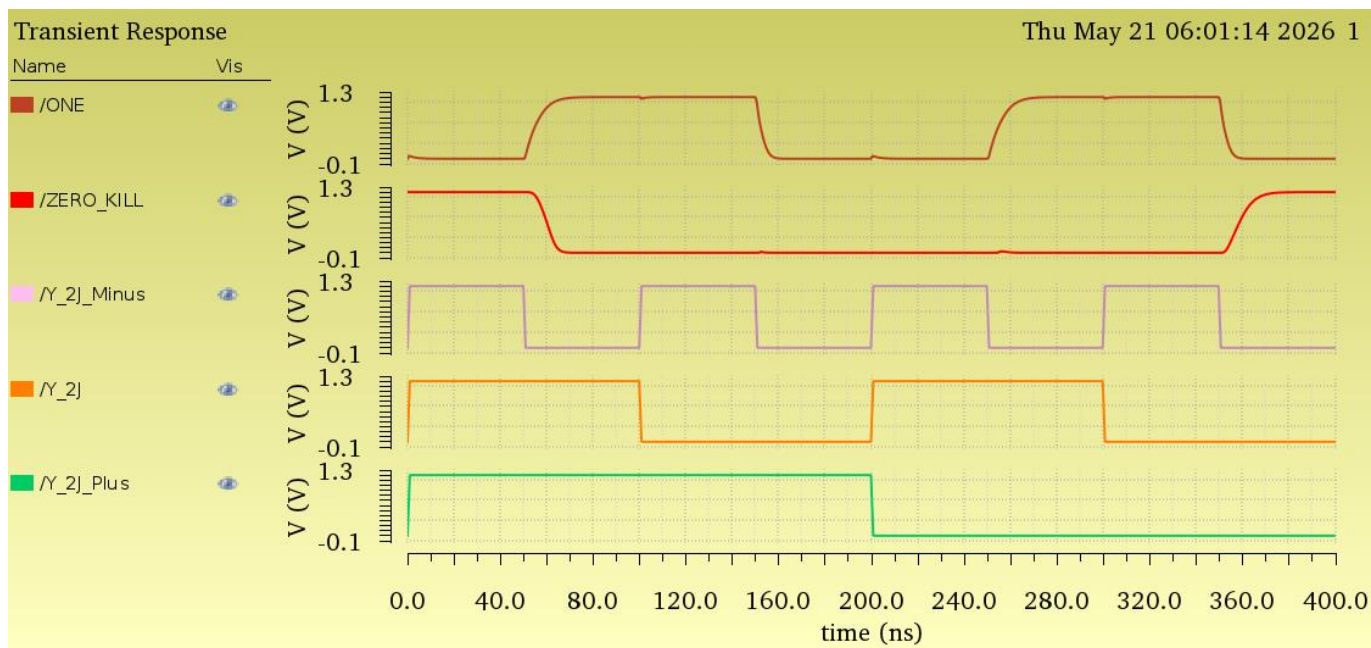


Figure 15: Booth encoder Test Bench pre-layout.

5. MUX

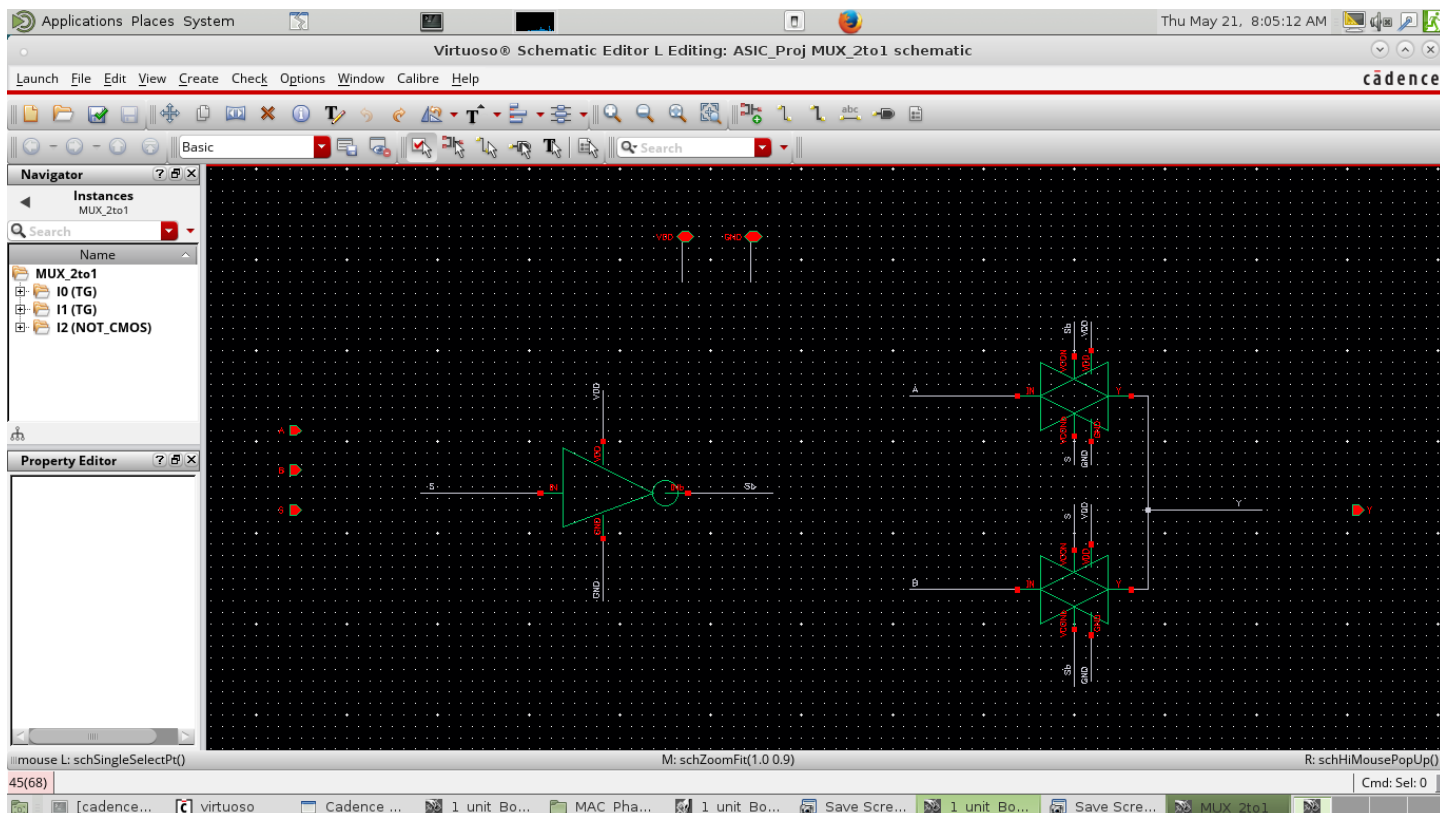


Figure 16: MUX Schematic.

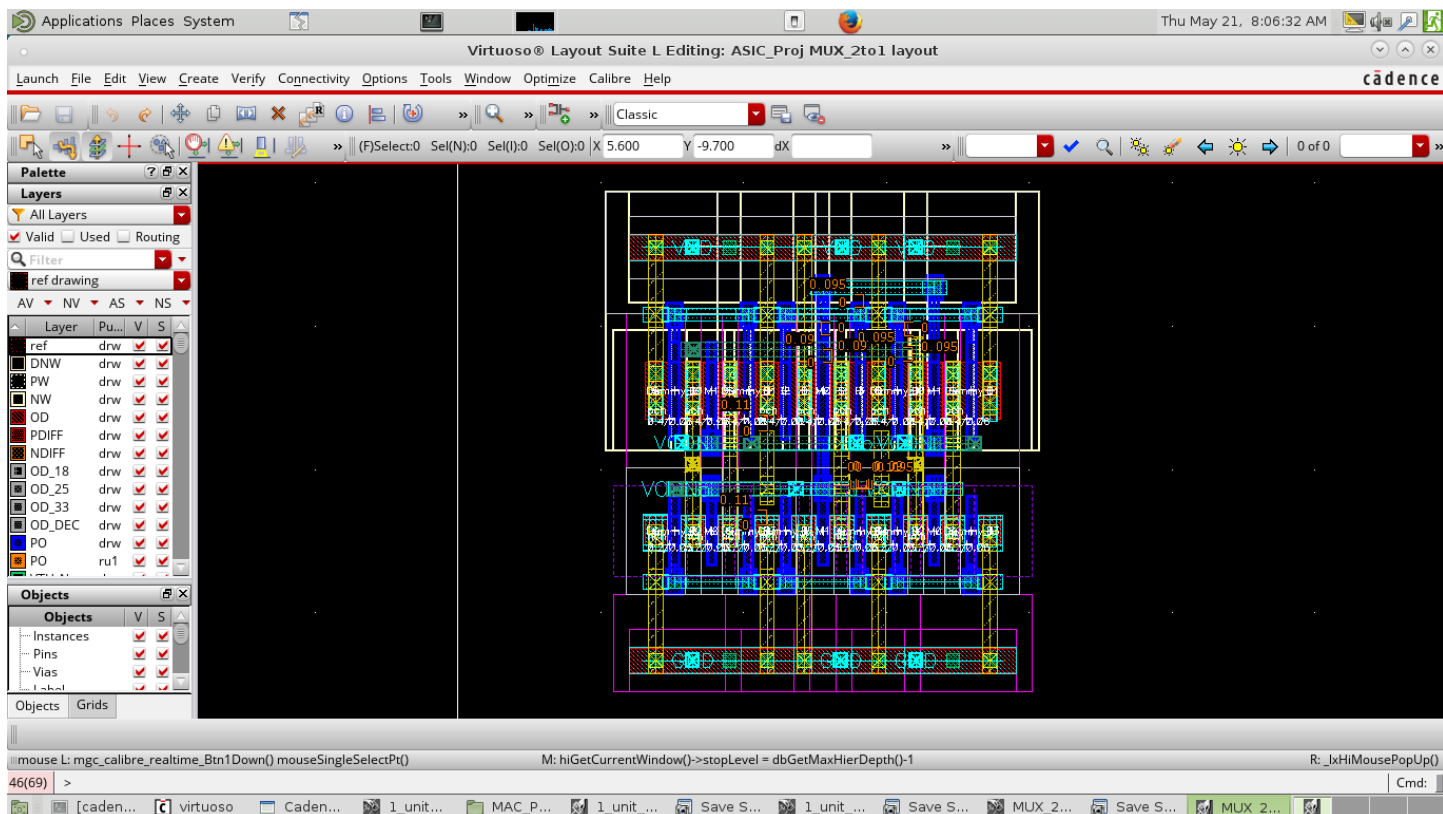


Figure 17: MUX Layout.

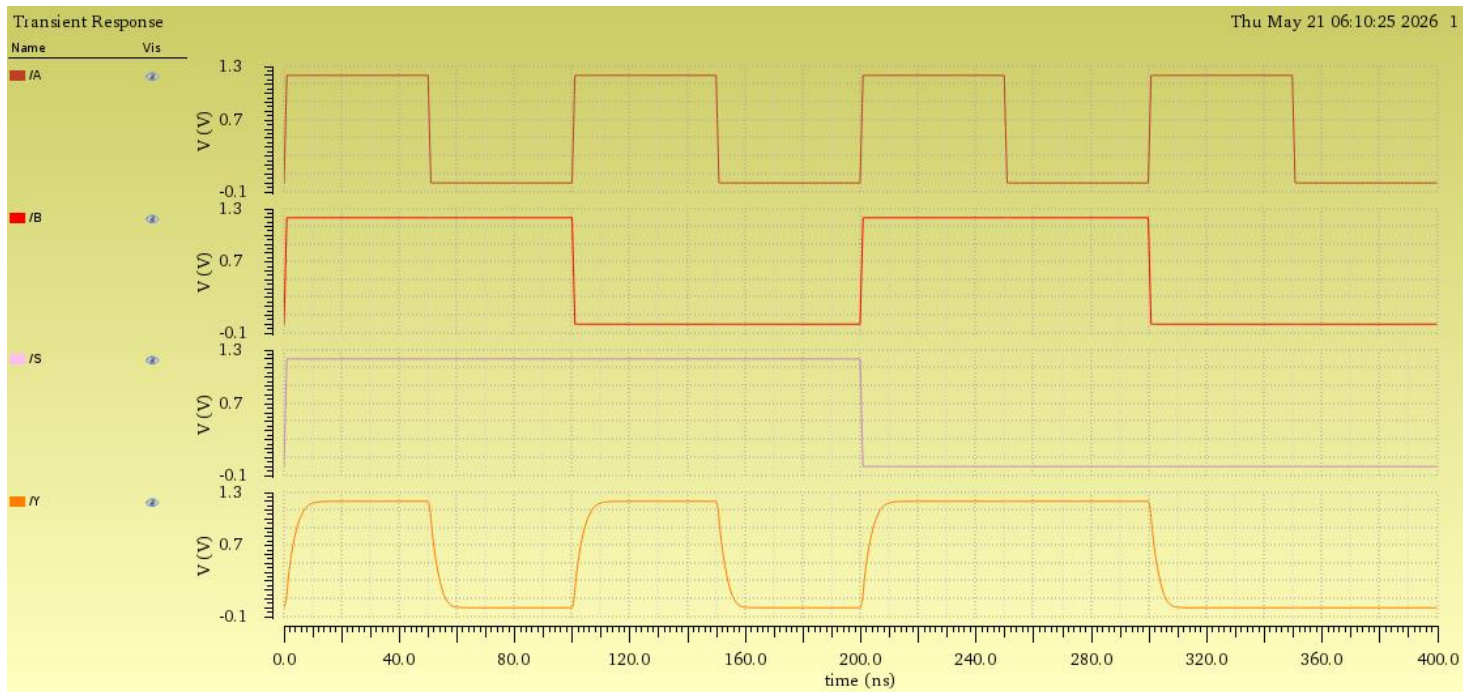


Figure 18: MUX Test Bench Pre-layout.

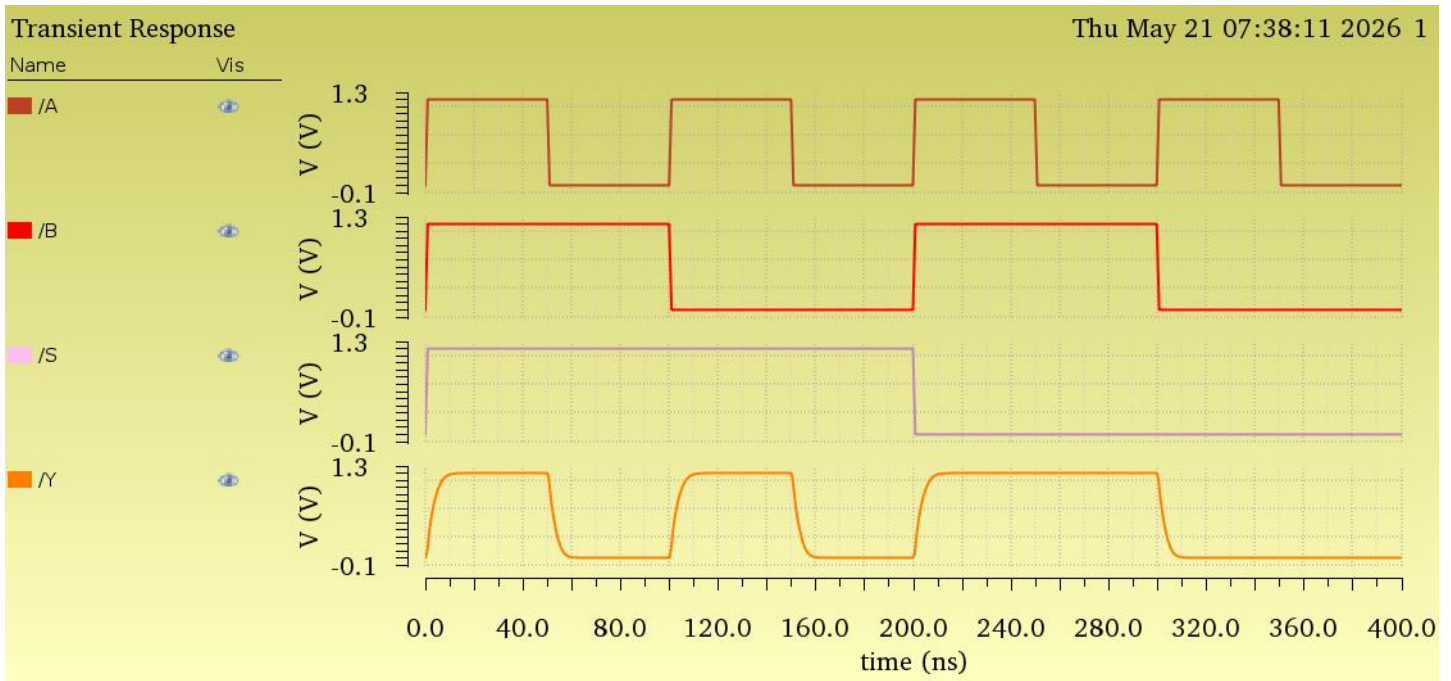


Figure 19: MUX Test Bench Post-Layout.

As we observe, the behaviour is identical pre- and post-layout.

6. C²MOS D flip flop

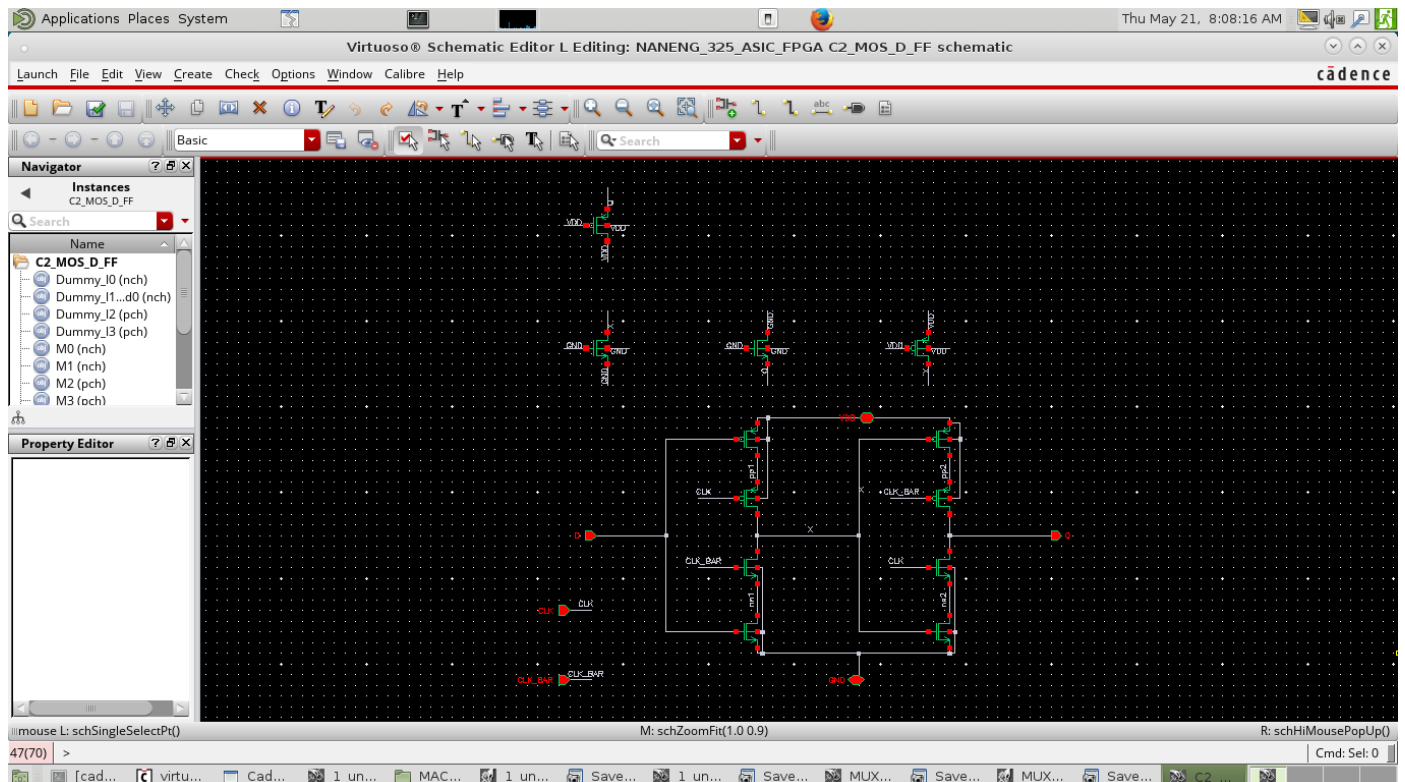


Figure 20: C²MOS D Flip Flop schematic.

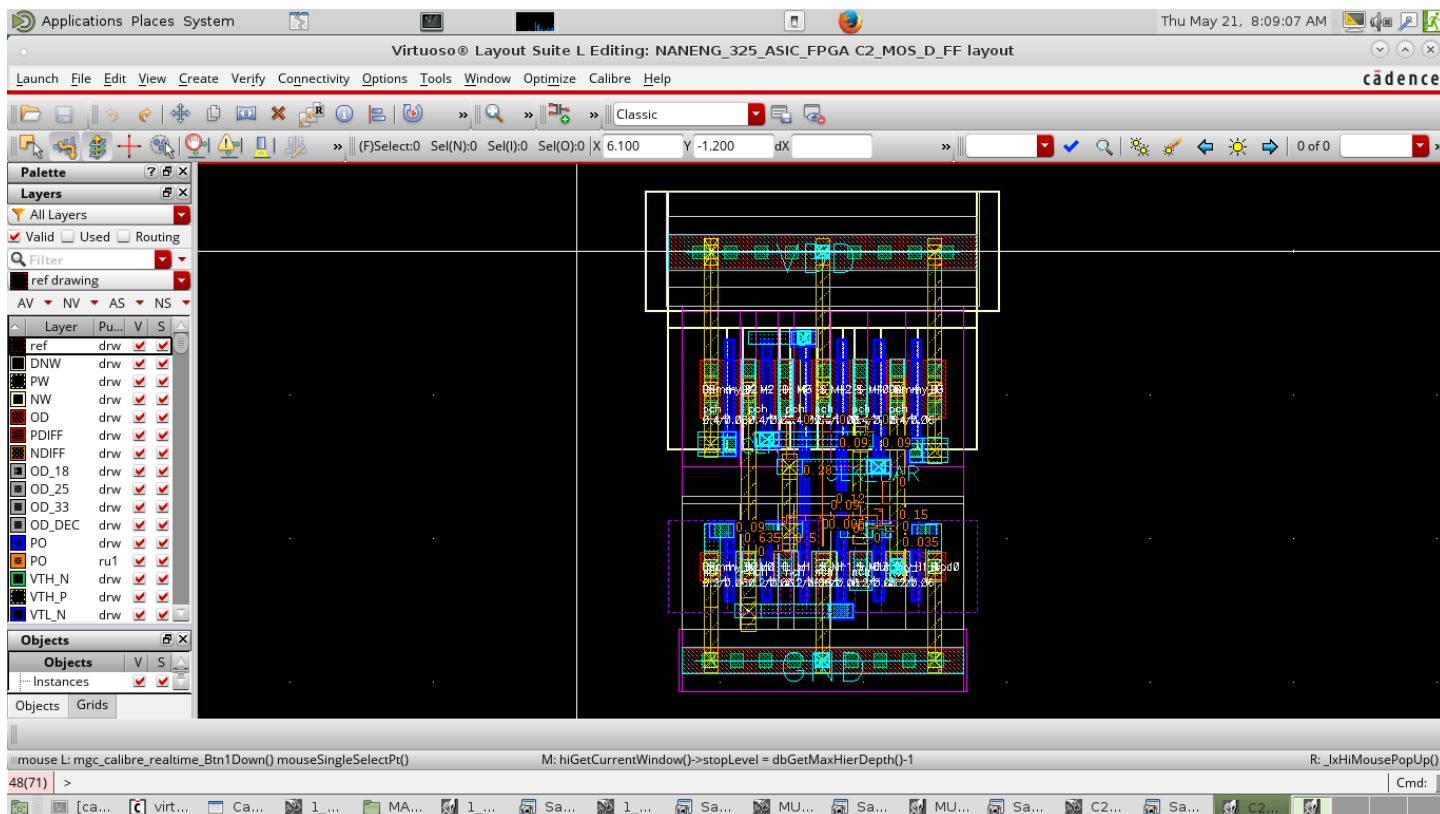


Figure 21: C²MOS Layout.

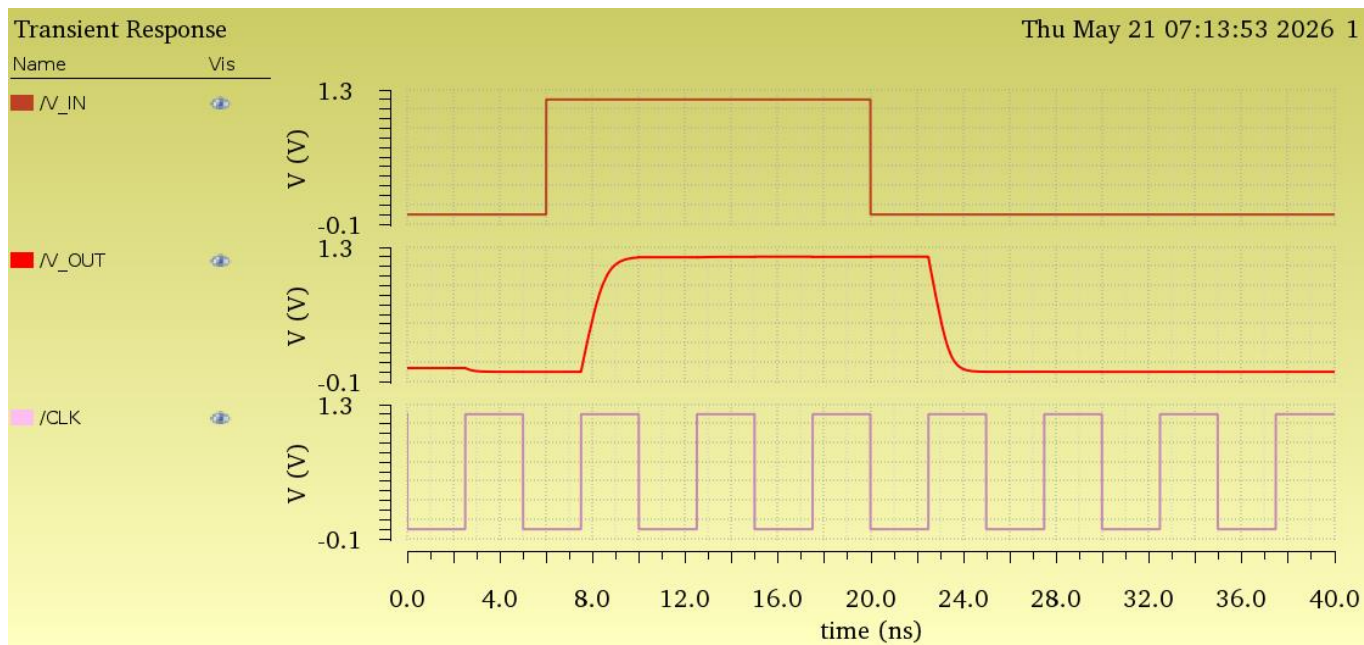


Figure 22: C²MOS flip flop Test Bench Pre-layout.

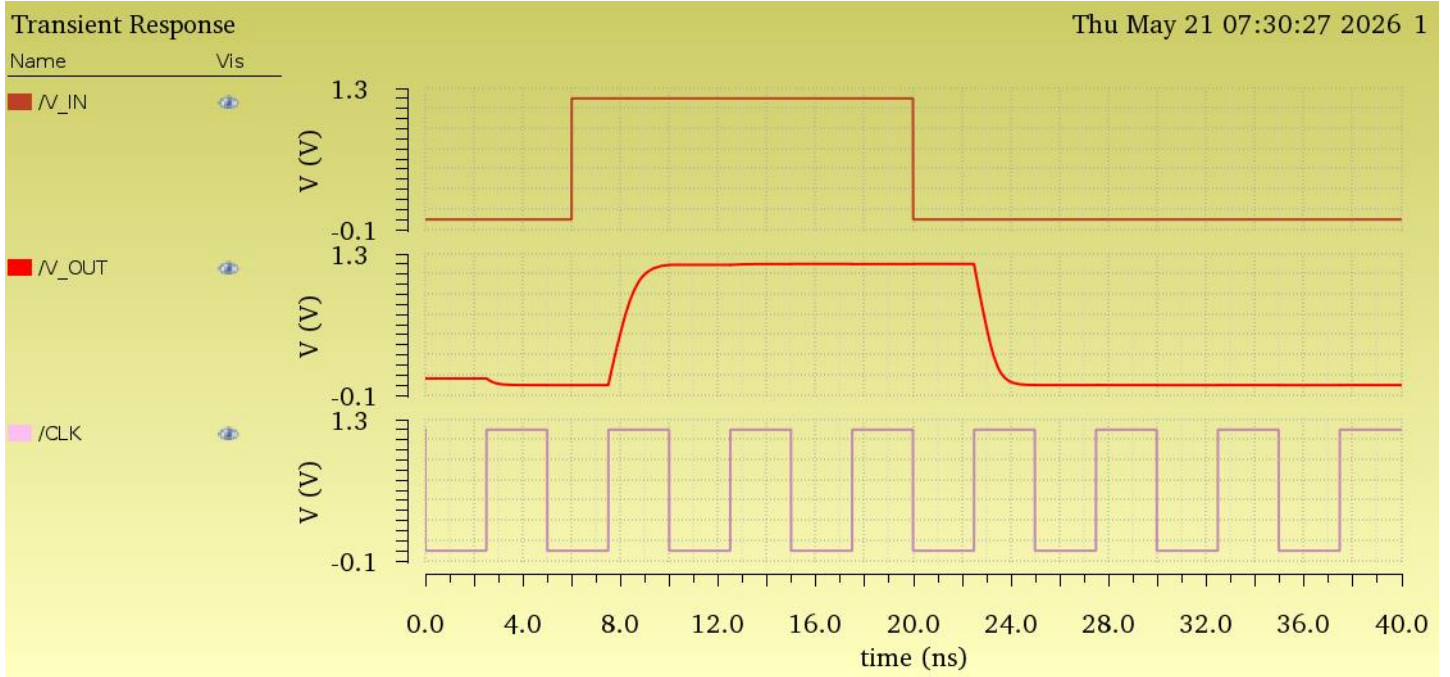


Figure 23: C^2 MOS flip flop Test Bench Pre-layout.

As we observe, the behaviour is identical pre- and post-layout.

7. Booth Encoder & Decoder

The Booth encoder and partial product generation units have been assembled together into one unit that takes as an input the multiplier and multiplicand and outputs all partial products to be added together in the Dadda tree.

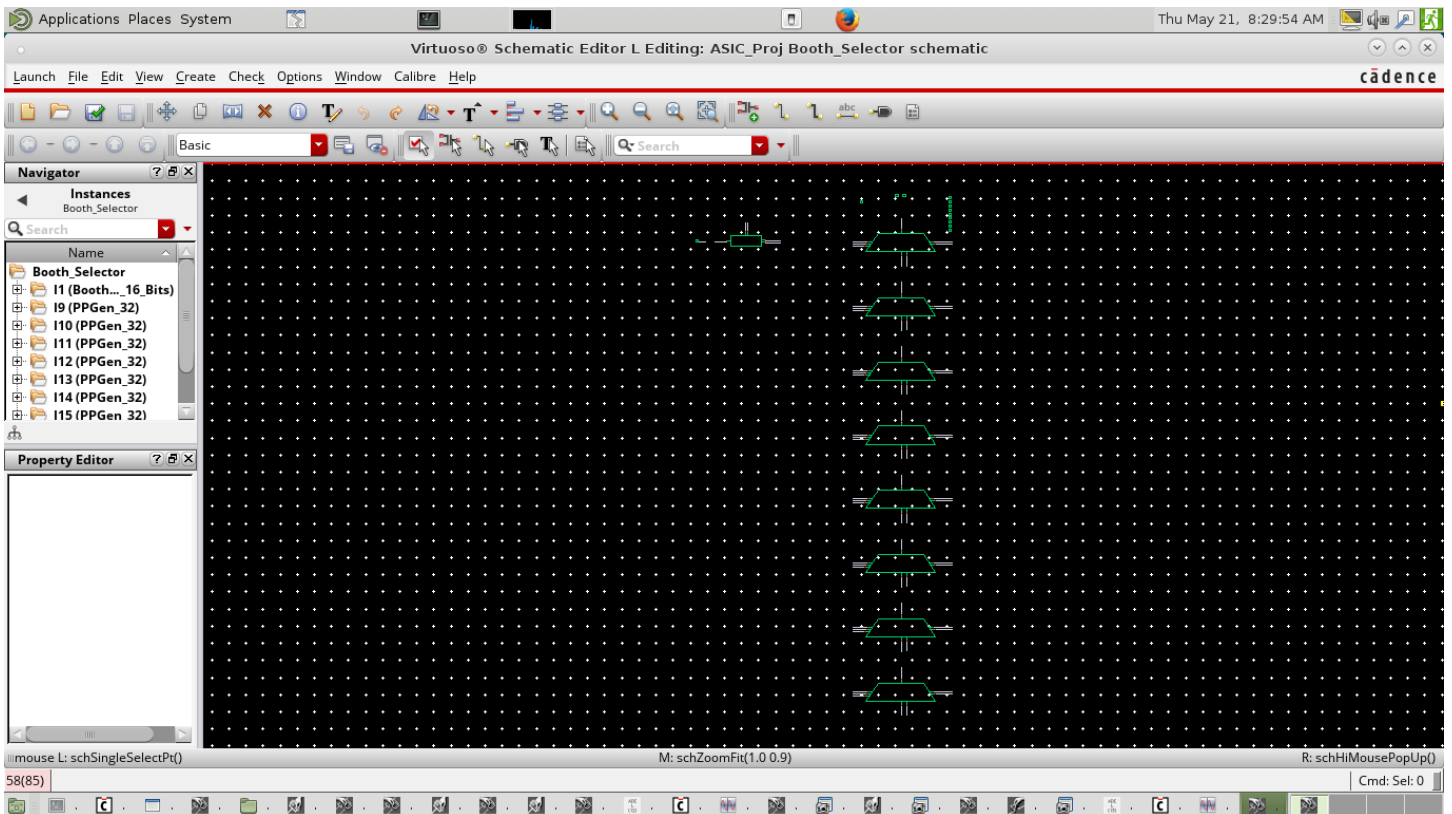


Figure 24: Booth Encoder Decoder Schematic.

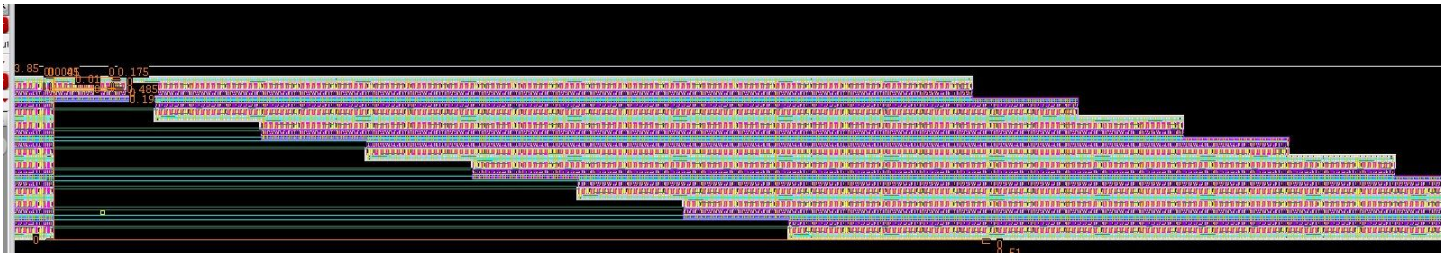


Figure 25: Booth Encoder Decoder Layout.

8. MAC Unit

The Booth encoder & decoder has been used inside the tree in order to create the MAC unit in one schematic.

Due to time insufficiency, we resorted to implementing an Booth multiplier or an array multiplier structure instead of a daddda tree structure. Its schematic is as depicted

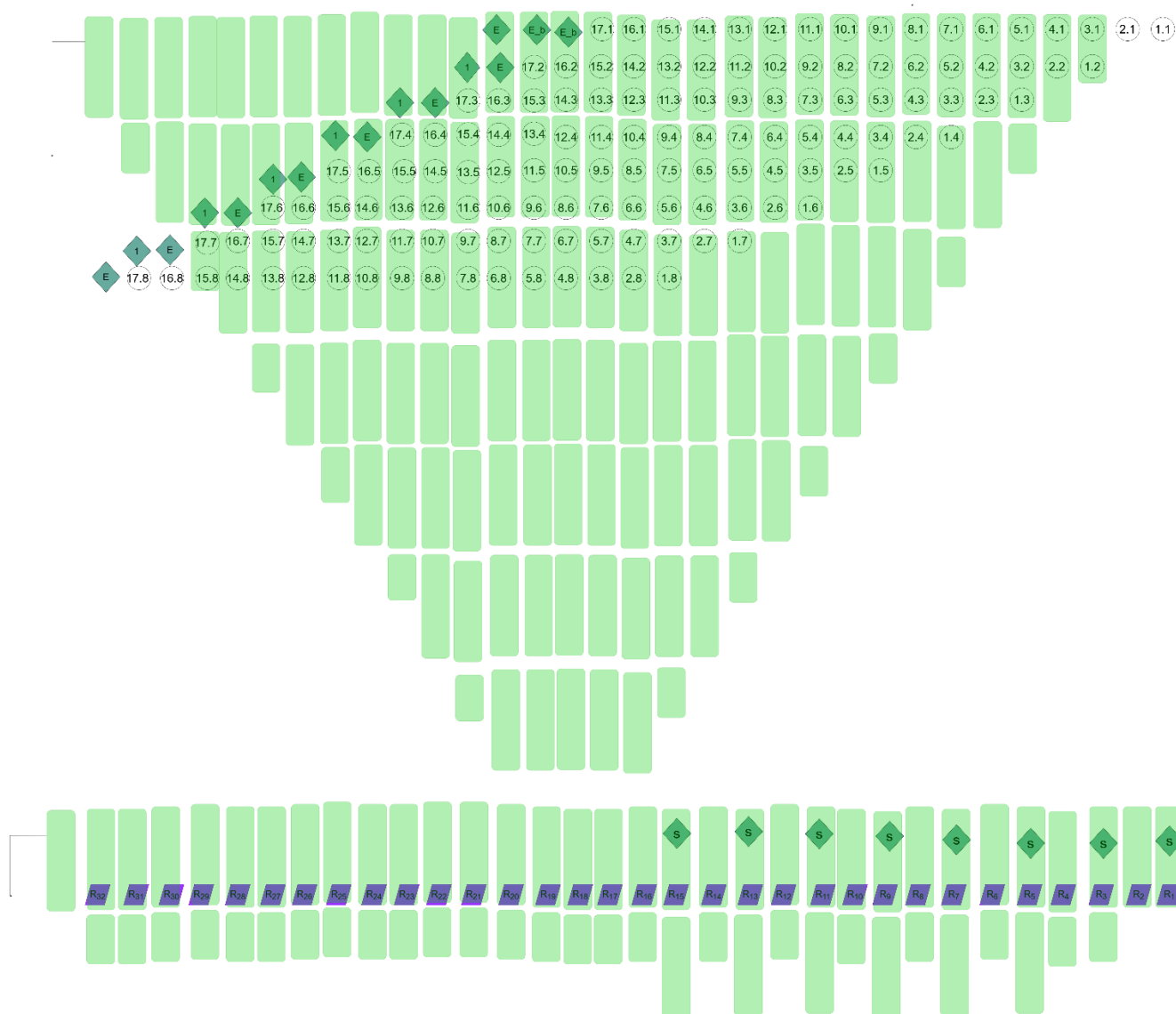


Figure 26: Schematic of the Booth Multiplier bit distribution and used Adders.

For a booth multiplier, there are S bits added to the beginning of each partial product (to implement negative partial products when adding 1 to the multiplicand's one's complement to create its two's complement). Thus, to improve tree regularity and symmetry, we decided to make a separate adder for these bits such that they are summed in parallel to the main summation network.

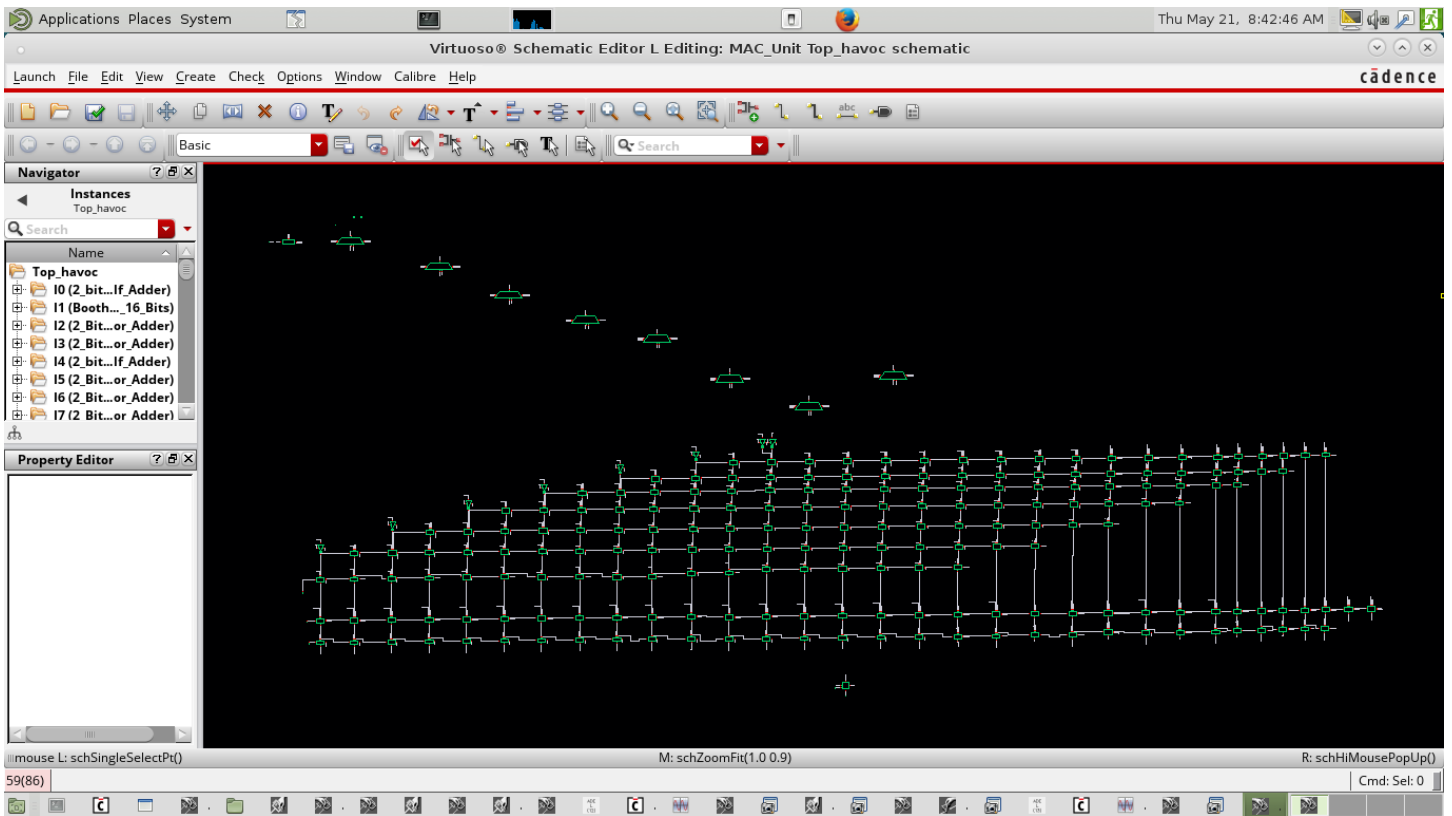


Figure 27: MAC Unit Schematic.

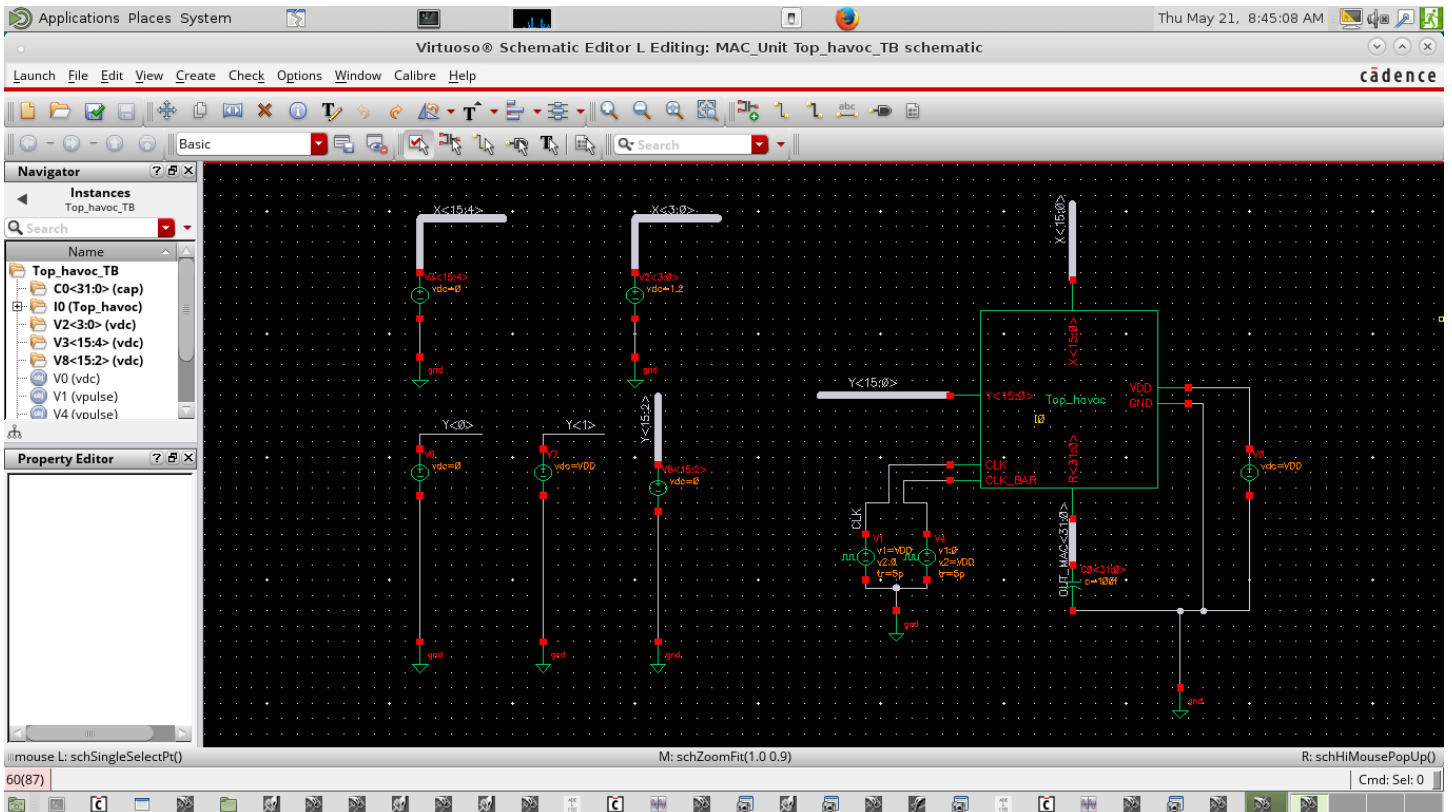


Figure 28: MAC Unit Test Bench.